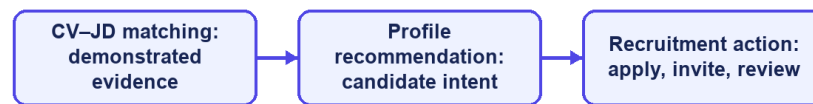


Three distinct recruitment concepts



CareerFit IT AutoPilot — thesis diagram

Figure 2.1. Distinction between matching, recommendation, and recruitment action

2.2 Text Representation and Preprocessing

2.2.1 Document Normalization

Before vectorization, text must be converted into a consistent representation. A typical pipeline extracts machine-readable text, normalizes character encoding and case, separates or standardizes tokens, removes formatting noise, and optionally filters stop words or applies stemming or lemmatization. In recruitment documents, preprocessing must preserve discriminative technical tokens such as framework names, programming languages, version-like expressions, and abbreviations. Aggressive removal can make a CV and a JD appear less related even when they share an important technology.

Bilingual data adds further complexity. Vietnamese contains diacritics and multi-syllable expressions separated by spaces, while English technical terminology frequently appears unchanged inside Vietnamese sentences. A practical pipeline must apply deterministic normalization to equivalent inputs and maintain a controlled vocabulary. It should also keep structured attributes—such as location, language, seniority, and salary mode—available for validation or filtering instead of forcing every business constraint into a single text vector.

Preprocessing quality directly affects every later score. Missing OCR text, malformed documents, extremely short descriptions, duplicated boilerplate, and

contradictory structured fields can distort term frequencies. For this reason, validation belongs before scoring. Hard validation rejects inputs that cannot support meaningful processing, while soft validation allows processing but records warnings or suggestions. This distinction prevents the ranking engine from silently assigning apparently precise scores to unusable data.

2.2.2 Bag-of-Words Representation

In a bag-of-words representation, a document is represented by the terms it contains and their weights; word order is not modeled directly. If the vocabulary contains $|V|$ terms, each document is mapped to a vector in $\mathbb{R}^{|V|}$. This representation is sparse because a particular CV or JD contains only a small portion of the complete vocabulary. Its main advantages are computational simplicity, reproducibility, and the ability to inspect which terms contribute to similarity. Its main limitation is lexical dependence: synonyms and related skills do not match unless normalization or an explicit mapping connects them.

2.3 TF-IDF and Cosine Similarity

2.3.1 Term Frequency and Inverse Document Frequency

The vector-space model represents documents and queries as weighted term vectors and ranks documents according to their geometric relationship [1]. TF-IDF is a standard weighting family for this model [2]. Term frequency (TF) reflects how strongly a term occurs in a particular document. Inverse document frequency (IDF) reduces the influence of terms that occur in many documents and increases the relative influence of terms that distinguish a smaller subset of the corpus.

Let $tf(t,d)$ denote the frequency of term t in document d , let N be the number of documents in the reference corpus, and let $df(t)$ be the number of documents containing t . A basic formulation is:

$$tfidf(t,d) = tf(t,d) \times \log(N / df(t)).$$

Implementations often use logarithmic TF scaling, smoothed IDF, or vector normalization to handle repeated terms and unseen values. The exact formulation changes numeric scores, so a thesis must document the implemented formula rather than referring to TF-IDF as if it were a single universal function. A controlled or static reference corpus can improve score stability because the IDF of a term does not

change whenever a user adds one document; however, it may become less representative as technologies and hiring terminology evolve.

2.3.2 Cosine Similarity

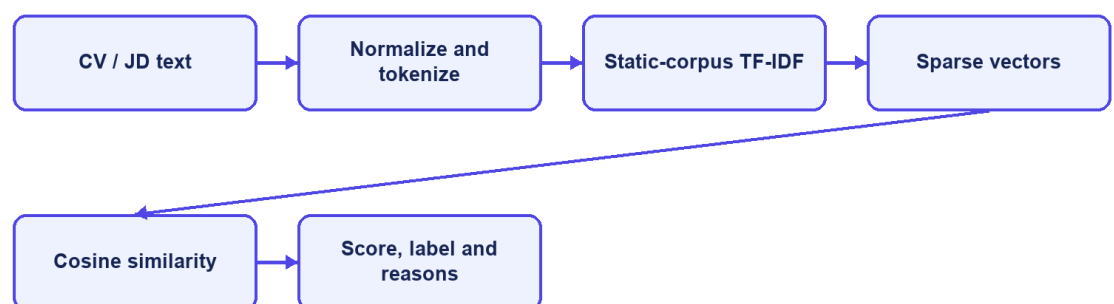
Cosine similarity compares the direction of two vectors rather than their unnormalized magnitude. For vectors x and y , it is defined as:

$$\cos(x,y) = (x \cdot y) / (||x||_2 ||y||_2).$$

For non-negative TF-IDF vectors, cosine similarity usually lies between 0 and 1. A larger value indicates a greater proportion of shared weighted terms. Length normalization is valuable for CV–JD comparison because a long CV should not receive a higher score merely because it contains more words. TF-IDF-weighted cosine distance has also been used as an efficient and reproducible document-alignment method [3].

Cosine similarity remains a lexical measure. If a JD uses “container orchestration” while a CV only uses “Kubernetes,” the vectors may not overlap unless the vocabulary or normalization layer establishes that relationship. Conversely, shared generic terms can increase similarity without proving that mandatory requirements are satisfied. CareerFit therefore treats similarity as one component of a broader workflow that also includes structured fields, validation, labels, reasons, user feedback, and policy conditions.

TF-IDF and cosine pipeline



CareerFit IT AutoPilot — thesis diagram

Figure 2.2. TF-IDF vectorization and cosine-similarity pipeline

2.4 Matching, Recommendation, and Ranking

Matching and recommendation can be formalized as ranking tasks. Given a query representation q and a candidate set D , the system computes a relevance score $s(q,d)$ for each $d \in D$ and orders the results by decreasing score. For candidate-to-job matching, q may be a CV vector and D the active job vectors. For recruiter-side discovery, q may be the selected JD and D the available candidate or CV vectors. For profile-based recommendation, q represents the candidate’s desired role, skills, location, seniority, and other preferences rather than one uploaded CV.

This separation matters because a CV describes demonstrated history, whereas a preference profile describes intent. A candidate may have experience in Java but seek a position involving Go; a CV-only ranking and a preference-based recommendation can therefore produce different, legitimate lists. Storing the workflows separately also makes evaluation more defensible: the relevance labels, candidate sets, and user objectives for one task should not be assumed to be valid for the other.

A raw similarity value is not automatically a calibrated probability of hiring success. Mapping it to a percentage or label is a presentation and business-rule decision, not a statistical guarantee. Thresholds such as Low, Medium, High, or Potential should be documented, tested at boundary values, and kept distinct from application status. Stable secondary sorting is also necessary when multiple items have equal scores so that repeated requests do not appear inconsistent.

Recent resume–job matching research increasingly uses dense encoders and contrastive learning. ConFit v2, for example, improves resume–job representations through hypothetical resumes and hard-negative mining [7]. Such approaches can capture semantic relations beyond exact terms, but they require training data, model governance, and an evaluation design appropriate to the domain. CareerFit intentionally begins with a transparent lexical baseline whose calculations can be reproduced and explained, while treating semantic or hybrid retrieval as future work rather than an implemented result.

2.5 Relevance Feedback and the Rocchio Method

2.5.1 Relevance Feedback

Relevance feedback uses judgments about retrieved items to modify a query or profile representation. Instead of assuming that the original text completely expresses the user's information need, the system observes which results are marked relevant or non-relevant. Explicit feedback is especially useful in recruitment because the importance of related skills may be known to the candidate or recruiter but omitted from the original CV, profile, or JD.

The classic Rocchio method updates a query vector by combining the original query with the centroids of relevant and non-relevant document vectors [4]. Let q_0 be the original vector, D_r the set of relevant feedback documents, and D_{nr} the set of non-relevant documents. A common form is:

$$q_m = \alpha q_0 + (\beta / |D_r|) \sum (d \in D_r) d - (\gamma / |D_{nr}|) \sum (d \in D_{nr}) d.$$

The parameters α , β , and γ control retention of the original representation, attraction toward relevant examples, and movement away from non-relevant examples. They are hyperparameters rather than universal constants. Positive and negative feedback may also be weighted differently because an explicit rejection can be ambiguous: a user may reject a job because of location, timing, or salary even when its technical content is relevant.

2.5.2 Feedback Semantics in CareerFit

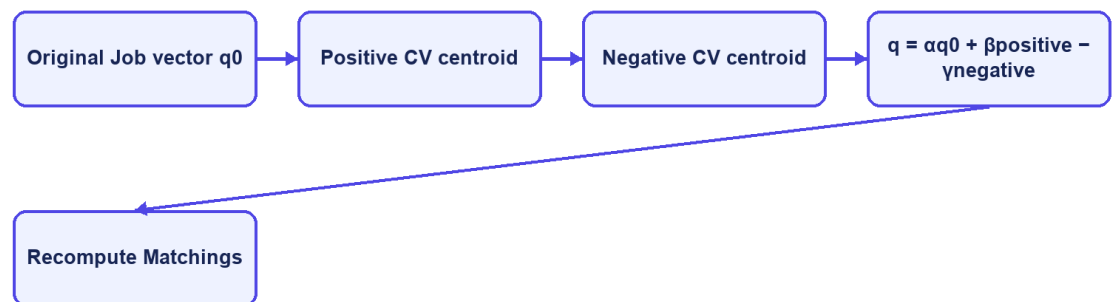
CareerFit distinguishes feedback such as GOOD_MATCH, POTENTIAL, BAD_MATCH, and NOT_INTERESTED. These labels should not be collapsed without a documented mapping. GOOD_MATCH provides a strong positive relevance signal. POTENTIAL may represent partial or transferable fit rather than direct relevance. BAD_MATCH is a negative judgment about the match. NOT_INTERESTED can reflect personal intent and may be weaker evidence about technical similarity. The implementation chapter must state exactly how each event affects the learned vector and whether role or channel changes its weight.

A robust update process retains the immutable base vector and recomputes the learned vector from the relevant feedback history. Repeatedly applying an update to the previously learned vector can create cumulative drift: running the same

recomputation more than once may change the output even though no new feedback exists. Idempotent recomputation means that the same base vector, feedback set, and parameters produce the same learned vector. This property is important for retries, scheduled processing, debugging, and auditability.

Feedback learning must be evaluated causally rather than by reporting only a final score. A controlled experiment records the baseline ranking, introduces a defined feedback event, recomputes the representation, and measures the effect on separate holdout items. If the positive item used to provide feedback is also the sole item used to claim improvement, the result is circular. Chapter 5 therefore separates feedback examples from holdout evaluation and explicitly labels synthetic scenarios.

Rocchio relevance feedback



CareerFit IT AutoPilot — thesis diagram

Figure 2.3. Rocchio relevance-feedback vector update

2.6 Evaluation Metrics for Ranked Results

Ranking quality cannot be adequately described by ordinary classification accuracy because users observe an ordered list and usually inspect only the first few items. Evaluation therefore requires relevance judgments and metrics that account for the cutoff position K , the location of relevant results, and, where available, graded relevance.

2.6.1 Precision@K, Recall@K, and HitRate@K

Precision@K is the proportion of the top K results that are relevant. Recall@K is the proportion of all known relevant items that appear in the top K. HitRate@K records whether at least one relevant item appears in the first K positions. Precision emphasizes the usefulness of the visible list, recall emphasizes coverage, and HitRate provides a simple task-success indicator. The metrics answer different questions and should not be substituted for one another.

2.6.2 Mean Reciprocal Rank

For a query, reciprocal rank is $1/r$, where r is the rank of the first relevant result; it is zero if no relevant result is retrieved. Mean Reciprocal Rank (MRR) averages this value across queries. MRR strongly rewards placing the first relevant result near the top but ignores the ordering of additional relevant items. It is therefore appropriate when finding one useful job or candidate is the primary objective, but it should be accompanied by a broader top-K metric when multiple relevant results matter.

2.6.3 Discounted Cumulative Gain

Discounted Cumulative Gain (DCG) supports graded relevance and discounts useful items that occur at lower ranks. Normalized DCG divides the observed DCG by the ideal ordering for the same relevance judgments, producing nDCG values that are comparable across queries. Järvelin and Kekäläinen introduced cumulated-gain measures to credit systems for ranking highly relevant documents before less relevant ones [5]. One common formulation is:

$$DCG@K = \sum_{i=1..K} (2^{rel_i} - 1) / \log_2(i + 1), \quad nDCG@K = DCG@K / IDC@K.$$

Metric values are meaningful only when the relevance labels, candidate set, cutoff K, and aggregation procedure are documented. A large improvement on a synthetic scenario demonstrates behavior under that scenario; it does not by itself demonstrate production hiring quality. Chapter 5 therefore reports dataset provenance, baseline construction, holdout logic, repeated-run behavior, and threats to validity together with the metric values.

2.7 Human-in-the-Loop and Explainable Automation

2.7.1 Levels of Human Involvement

Human-in-the-Loop is not a single interface feature. Human involvement can occur during data validation, model configuration, review of recommendations, approval of actions, exception handling, and post-action feedback. NIST describes human–AI configurations as ranging from manual decision making through systems that provide an additional opinion or defer to an expert, to more autonomous operation [9]. The appropriate configuration depends on the consequences and failure modes of the task.

Recruitment decisions can materially affect people, so a similarity engine should assist prioritization rather than act as an unquestioned authority. CareerFit separates perception, decision support, action, learning, and audit. The matching engine produces evidence; the AutoFit policy evaluates whether an action is permitted; a user may review or confirm important actions; feedback changes later ranking; and audit records support investigation. This design does not eliminate bias, but it makes the control points and system state more explicit.

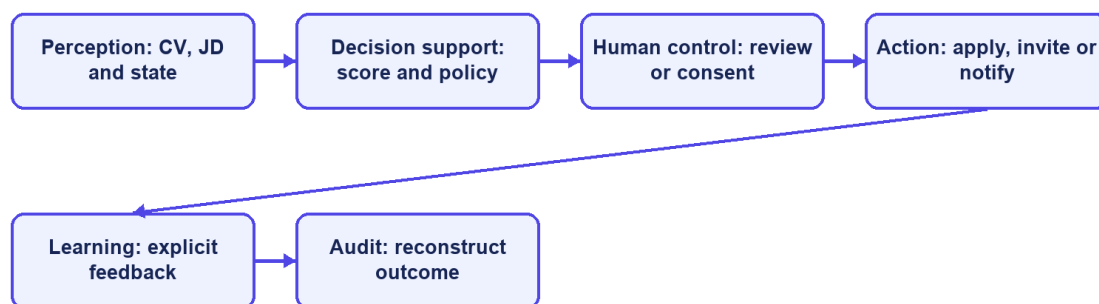
2.7.2 Explainability and Auditability

Explainability answers why a particular result or action was produced, whereas auditability concerns whether the inputs, rules, actor, channel, time, and outcome can be reconstructed. A list of overlapping skills can provide a local explanation of lexical similarity, but it is not proof that the entire model or business decision is fair. Likewise, an audit log records what occurred but does not guarantee that the recorded policy was appropriate.

Research on algorithmic hiring warns that claims about bias mitigation and performance must be evaluated against actual practices, data, and institutional context [10]. Explanations can support understanding, but they should be grounded in the features used by the system. Work on explainable recommendation similarly distinguishes recommendation scoring from the generation of user-facing reasons and emphasizes fact-grounded explanations [14]. CareerFit therefore favors inspectable match reasons and policy outcomes over unsupported natural-language claims about candidate suitability.

The NIST AI Risk Management Framework organizes AI risk work around governance, mapping, measurement, and management [9]. CareerFit is not claimed to be a complete implementation of that framework; however, its design aligns with several practical principles: document scope and limitations, preserve human authority for consequential actions, measure behavior using defined evidence, and maintain records that support review.

Human-in-the-Loop cycle



CareerFit IT AutoPilot — thesis diagram

Figure 2.4. Human-in-the-Loop control cycle in CareerFit

2.8 Web Architecture, Security, and Audit Foundations

2.8.1 Client–Server and REST

CareerFit uses a web client and a server-side application connected through HTTP APIs. The client is responsible for presentation, navigation, local interaction state, and invoking authorized operations. The backend enforces authentication, authorization, validation, business rules, transactions, persistence, and background processing. This separation prevents browser-side presentation logic from becoming the authority for protected actions.

REST is an architectural style characterized by constraints including client–server separation, stateless interaction, cacheability where appropriate, a uniform interface, layered components, and optional code-on-demand [11]. A REST API models resources and transfers representations of their state. In practice, merely using JSON over HTTP does not guarantee that every REST constraint is satisfied; Chapter

4 therefore describes CareerFit as a REST-oriented API and documents its actual endpoint and state-transition behavior.

2.8.2 Authentication, Authorization, and Action Tokens

Authentication establishes the identity associated with a request, while authorization determines whether that identity may perform an operation. Role-Based Access Control assigns permissions according to roles such as Candidate, Recruiter, or Administrator, but ownership checks are still required; for example, one candidate must not access another candidate's private CV merely because both share the same role.

JSON Web Token is a compact, URL-safe format for transferring claims that can be signed or message-authentication-code protected and may include registered claims such as subject, audience, expiration time, issued-at time, and token identifier [12]. A valid signature alone is not sufficient: the application must validate the expected algorithm and relevant claims and must apply its own account and authorization rules. Short-lived access tokens and purpose-specific action tokens address different workflows and should not be treated as interchangeable credentials.

An actionable email link can be opened by the intended user, forwarded accidentally, or visited automatically by a mail-security scanner. A safer target pattern lets a GET request display a confirmation page without changing business state and requires a deliberate POST request to execute the action; the current CareerFit email-action implementation does not yet satisfy this target. The action token should be random or cryptographically protected, purpose-bound, expiring, and single-use. Server-side consumption state or an equivalent replay-prevention mechanism is needed because expiration alone does not prevent the same valid link from being used repeatedly.